CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 11: Inequalities

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In triangle ABC, $\angle A = 80^\circ$ and $\angle B = 40^\circ$. Name the longest and the shortest sides.
2. In triangle PQR, $PQ = 4$ cm, $QR = 6$ cm, $RP = 5$ cm. Arrange the angles in descending order.
3. Can 7 cm, 8 cm and 15 cm be the sides of a triangle?
4. Can 5 cm, 6 cm and 10 cm be the sides of a triangle?
5. Two sides of a triangle are 8 cm and 13 cm. Between what limits does the third side lie?
6. In a right-angled triangle, which side is the longest, and why?
7. In triangle ABC, $AB = AC$ and $\angle A = 100^\circ$. Which is longer, AB or BC?
8. In triangle ABC, the exterior angle at C is 100° and $\angle A = 30^\circ$. Arrange the sides in ascending order.
9. The sides of a triangle are 3 cm and 7 cm, and the third side is a whole number. What are the possible values?
10. P is a point 6 cm from a line. Can a segment from P to the line have length 5 cm?

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Answer Key

1. Longest BC, shortest AC
2. $\angle P > \angle Q > \angle R$
3. No, since $7 + 8 = 15$
4. Yes, since $5 + 6 > 10$
5. Between 5 cm and 21 cm
6. The hypotenuse; it is opposite the right angle, which is the largest angle
7. BC, since $\angle A = 100^\circ$ is greater than $\angle C = 40^\circ$
8. Angles are $30^\circ, 70^\circ, 80^\circ$ at A, B, C; so $BC < AC < AB$
9. 5, 6, 7, 8, 9 cm
10. No; the perpendicular (6 cm) is the shortest segment

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